

Class 11th Biology (Botany & Zoology) Syllabus

Theory Marks: 70 | Practical Marks: 30 | Total Marks: 100 | Duration: 3 Hours

Section A: Botany (35 Marks)

Unit I: Diversity of Life (9 Marks)

- Variety of living organisms; Need and history of classification (Artificial, Natural and Phylogenetic).
- Biosystematics: Taxonomy and Systematics; Concept of species & Taxonomical hierarchy; Binomial nomenclature; Herbarium.
- Two Kingdom & Five Kingdom classifications, merits and demerits.
- General characters & classification of Monera, Protista and Fungi; Lichens; Status of Viruses and Viroids.

Unit II: Kingdom Plantae & Plant Morphology (9 Marks)

- Classification of plants into major groups; General characters of Algae, Bryophyta, Pteridophyta and Gymnosperms.
- Morphology of Flowering plants and functions: Root, stem, leaf (without modifications).
- Morphology of Inflorescence, flower, fruit and seed; Detailed description of family Solanaceae.

Unit III: Plant Anatomy (5 Marks)

- Tissue systems in plants: Epidermal, ground and vascular tissue systems.
- Anatomy and functions of dicot and monocot root, stem and leaves.

Unit IV: Plant Physiology & Development (12 Marks)

- Respiration: Cellular respiration, Glycolysis, Krebs's cycle, Electron Transport System, ATP energetics, Respiratory Quotient.
- Photosynthesis: Pigments, Light reaction (Photosystems I & II, cyclic/non-cyclic photophosphorylation), Dark reaction (C3 & C4 cycles), Photorespiration.
- Plant Growth & Regulators: Auxins, Gibberellins, Cytokinins, Ethylene, Abscissic Acid (ABA).

Section B: Zoology (35 Marks)

Unit I: Diversity in Living World (Animal Kingdom) (8 Marks)

- General characters and classification of animals (Non-chordates up to phyla, chordates up to class level).
- National Parks with special reference to J&K;: Dachigam, Kishtwar, Salim Ali, Kazinag and Hemis.
- Sanctuaries and Biosphere reserves.

Unit II: Structural Organisation & Animal Biomolecules (7 Marks)

- Morphology, Anatomy and organ systems of Frog (digestive, circulatory, respiratory, nervous, reproductive).
- Biomolecules: Structure and function of Carbohydrates, Proteins, Lipids, Nucleic Acids; Primary & secondary metabolites.
- Enzymes: Types, properties, mechanism of action and factors affecting.

Unit III: Cell Structure and Function (8 Marks)

- Cell theory; Prokaryotic and Eukaryotic cell structure; Cell wall, Cell membrane.
- Cell Organelles: Mitochondria, ER, Golgi, Ribosomes, Lysosomes, Plastids, Nucleus, Cytoskeleton, Cilia & Flagella.
- Cell Division: Cell cycle phases, Mitosis and Meiosis with significance.

Unit IV: Human Physiology (12 Marks)

- Breathing & Respiration: Respiratory organs, mechanism of breathing, gas exchange and transport, respiratory disorders.
- Body Fluids & Circulation: Blood groups, Rh factor, Lymph, Cardiac cycle, ECG, double circulation, disorders.
- Excretory Products & Elimination: Nephron function, urine formation, osmoregulation, kidney regulation.
- Locomotion & Movement: Muscle contraction mechanism, skeletal framework, joints.
- Neural & Chemical Coordination: Neuron impulse conduction, Central nervous system, Endocrine hormones and disorders.

Prescribed Textbooks:

- A Textbook of Biology for Class XI published by NCERT / JKBOSE